Junie: Your AI Coding Agent in JetBrains IDEs

Autonomous Coding with Safety Controls

Transform Your Development Workflow
with AI-Powered Automation

Four-Hour Hands-On Workshop

🎯 What You'll Master Today

Core Concepts

- 🔍 **Ask Mode**: Analysis & insights
- ⚡ **Code Mode**: Automated changes
- 📋 **Guidelines**: Team conventions
- 🛡️ **Safety**: Approvals & controls

Hands-On Labs

- ☕ **Java**: Spring Boot APIs
- 🐍 **Python**: Refactoring & testing
- ⚛️ **React**: Forms & validation
- 🔧 **MCP**: External tool integration

📚 Prerequisites Check

✅ **JetBrains IDE** (IntelliJ IDEA, PyCharm, WebStorm, GoLand, PhpStorm, RustRover, RubyMine)

✅ **JetBrains AI** subscription (Free tier available!)

✅ **Development Environment** (Java 17+ / Python 3.8+ / Node 16+)

🆓 Free Tier: Credit-based cloud assistance perfect for getting started

💡 Quick Setup: We'll install Junie together in the first 5 minutes

🚀 Installation in 60 Seconds

Open IDE Settings → Plugins Marketplace → Search Junie → Install & Restart → View Tool Windows

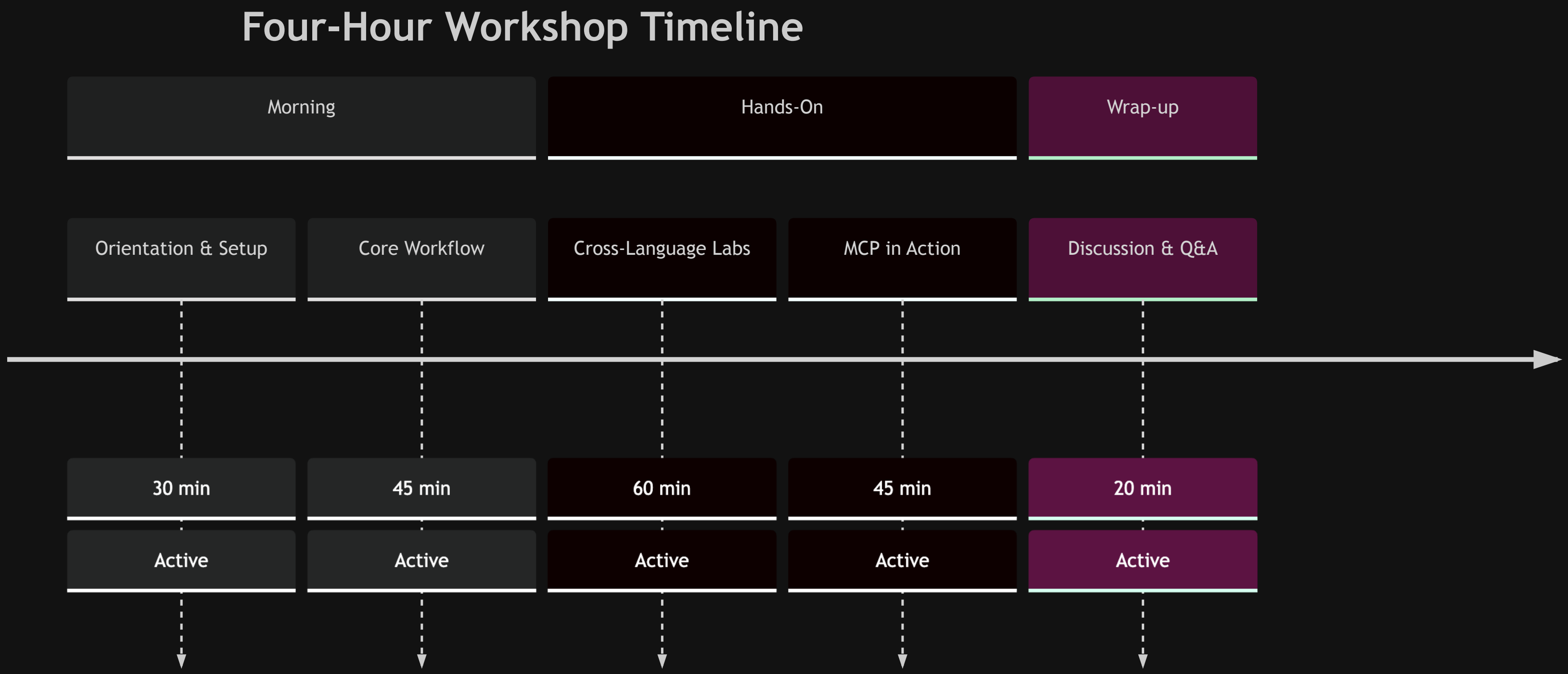
Available Models:

🎯 GPT-5

🧠 Claude 3.5 Sonnet

⚡ Claude 4.5 Sonnet

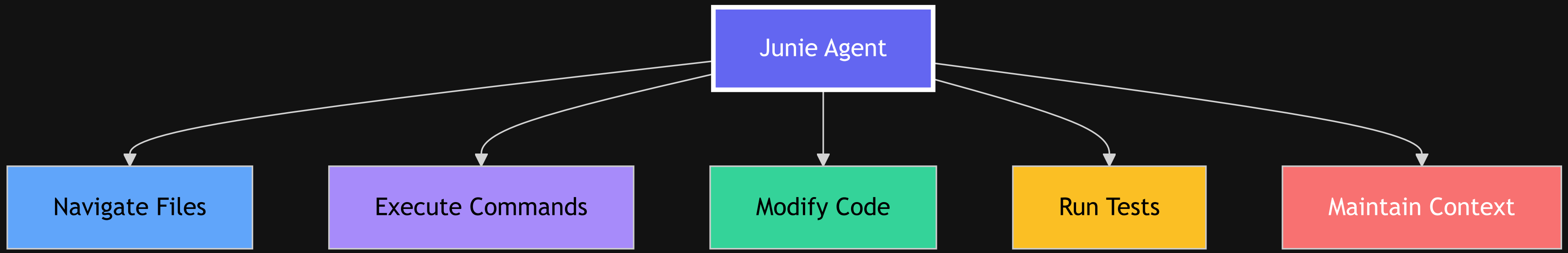
📅 Today’s Journey



☕ Break after Core Workflow segment

🤖 What is Junie?

JetBrains' AI-Powered Coding Agent



🔄 Two Operating Modes

🔍 Ask Mode

Read-Only Analysis

- Explain complex code
- Analyze architecture
- Find bugs & issues
- Review security
- Check test coverage

Perfect for understanding before changing

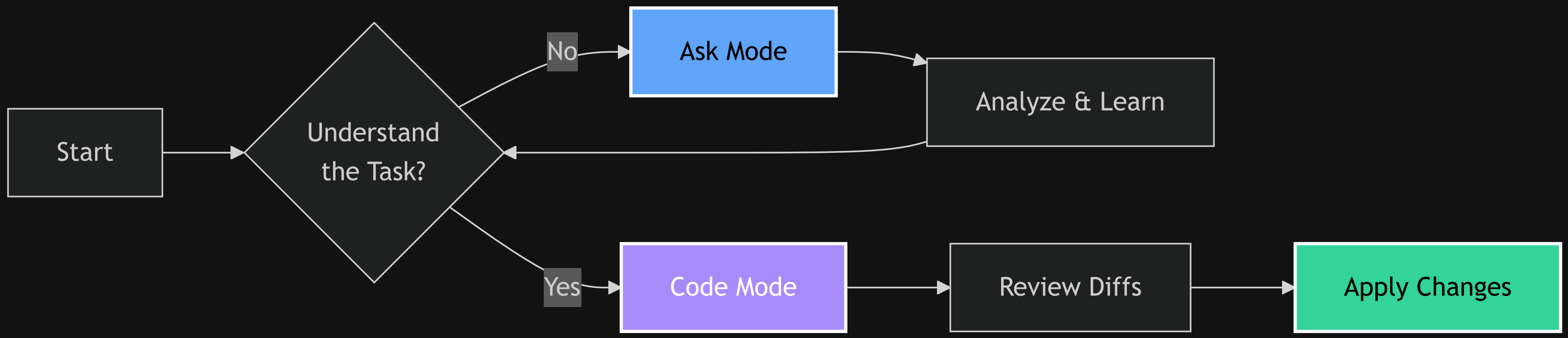
⚡ Code Mode

Make Changes

- Implement features
- Fix bugs
- Refactor code
- Generate tests
- Update dependencies

Executes multi-step plans with diffs

💡 Pro Workflow



✨ Always start with Ask mode for complex tasks

📝 Project Guidelines

Location: `.junie/guidelines.md`

```
## Technology Stack
- Framework: Spring Boot 3.2
- Testing: JUnit 5 + AssertJ

## Conventions
- REST: /api/v1/{resource}
- DTOs: Java records
- Tests: Given-When-Then
```

❌ Without Guidelines

Inconsistent patterns

Multiple review cycles

Style conflicts

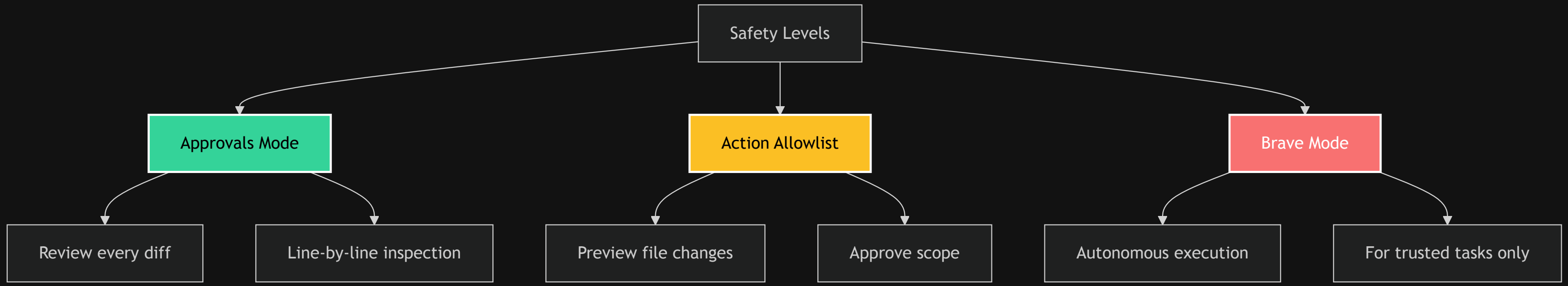
✅ With Guidelines

Consistent output

70% fewer reviews

Team alignment

Safety Controls



⚡ When to Use Each Mode

Mode	Good For	Not For
Approvals	• Critical business logic • First-time tasks • Learning Junie	• Repetitive formatting • Simple refactors
Allowlist	• Known file sets • Defined scope • Team reviews	• Exploratory changes • Unknown impact
Brave	• Test generation • Formatting • Documentation	• Production code • Database changes • First attempts

🔌 MCP: Model Context Protocol

Available Tools

📚 context7

- Real-time library docs
- 1000+ libraries
- Version compatibility
- Optional API key (higher rate limits)

🎭 Playwright

- E2E test generation
- Browser automation
- Accessibility-first
- TypeScript output

Configuration

```
{
  "mcpServers": {
    "context7": {
      "command": "npx",
      "args": ["-y", "@upstash/context7-mcp"]
    },
    "playwright": {
      "command": "npx",
      "args": ["@playwright/mcp-server"]
    }
  }
}
```

Settings → Junie → MCP → Edit mcp.json

🧪 Lab Overview

Lab A: Java → Lab B: Python → Lab C: React → Lab D: Upgrades → Demo: Playwright

Timing:

- Lab A: 45-60 min
- Lab B: 30-45 min
- Lab C: 30-45 min
- Lab D: 20-30 min
- Demo: 20-30 min

☕ Java
Spring Boot
REST APIs
JUnit + AssertJ

🐍 Python
PEP 8
Type hints
pytest

⚛️ React
TypeScript
Forms
Testing Library

🔧 MCP
context7
Upgrades
Migration

☕ Lab A: Spring Boot Journey

Analyze → Create API → Review Diffs → Generate Tests → Add Guidelines → Regenerate → Consistent Output

Without Guidelines

- Inconsistent patterns
- Various styles
- More review cycles

With Guidelines

- Consistent code
- Team standards
- Faster approval

🐍 Lab B: Python Transformation

Before 😱

```python
def calc(x,y,op):
    if op=="add": return x+y
    elif op=="sub": return x-y
```

- No type hints
- Poor naming
- No tests
- No docs

After ✨

```python
def calculate(
    x: float,
    y: float,
    operation: str
) -> Optional[float]:
    """Calculate result.

    Args:
        x: First number
        y: Second number
        operation: Operation

    Returns:
        Calculation result
    """
```

⚛️ Lab C: React Forms

```
interface RegistrationForm {
  email: string;       // valid email
  password: string;    // min 8, special char
  confirm: string;     // must match
  terms: boolean;      // required
}
```

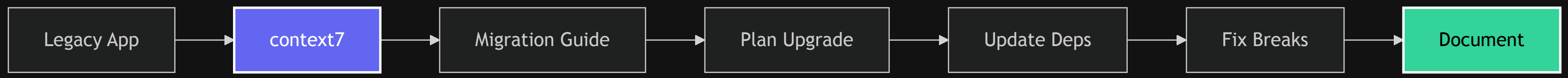
React Hook Form
Form state

Zod
Validation

Testing Library
90% coverage

Accessibility
ARIA compliant

🔧 Lab D: Smart Upgrades with context7



React 17 → React 18

Legacy Modern

🎭 Demo: Playwright Magic

Generate E2E Tests in 5 Minutes

```
test('successful login', async ({ page }) => {
  const loginPage = new LoginPage(page);
  await loginPage.goto();
  await loginPage.login('user@example.com', 'pass');

  await expect(page).toHaveURL('/dashboard');
  await expect(page.getByRole('heading', { name: 'Welcome' }))
    .toBeVisible();
});
```

✅ Page Objects

✅ Accessibility

✅ TypeScript

✅ Retry Logic

⚔️ Junie vs The Competition

Feature	Junie	Copilot	Cursor	Codeium
Multi-file edits	✅ **Full**	❌	✅	⚠️
Test execution	✅ **Yes**	❌	❌	❌
Rollback	✅ **Yes**	❌	⚠️	❌
MCP tools	✅ **Yes**	❌	❌	❌
IDE native	✅ **Full**	⚠️	❌	⚠️

🎯 Common Patterns

TDD Flow

1. Ask: "What tests?"
2. Code: "Generate tests"
3. Code: "Implement"

Red → Green → Refactor

Safe Refactoring

1. Generate tests
2. Identify issues
3. Refactor safely

Tests stay green

Upgrades

1. context7 check
2. Update deps
3. Fix breaks

Systematic migration

🚀 CI/CD Integration

```yaml
name: Junie Guidelines Check
on: [pull_request]

jobs:
  guidelines-compliance:
    runs-on: ubuntu-latest
    steps:
      - uses: actions/checkout@v3
      - name: Check guidelines
        run: |
          # Validate against .junie/guidelines.md
          # Ensure consistency
```

Pre-commit
Local validation

CI Pipeline
Automated checks

PR Review
Team alignment

✨ Key Takeaways

✅ **Ask before Code** - Understand first, implement second

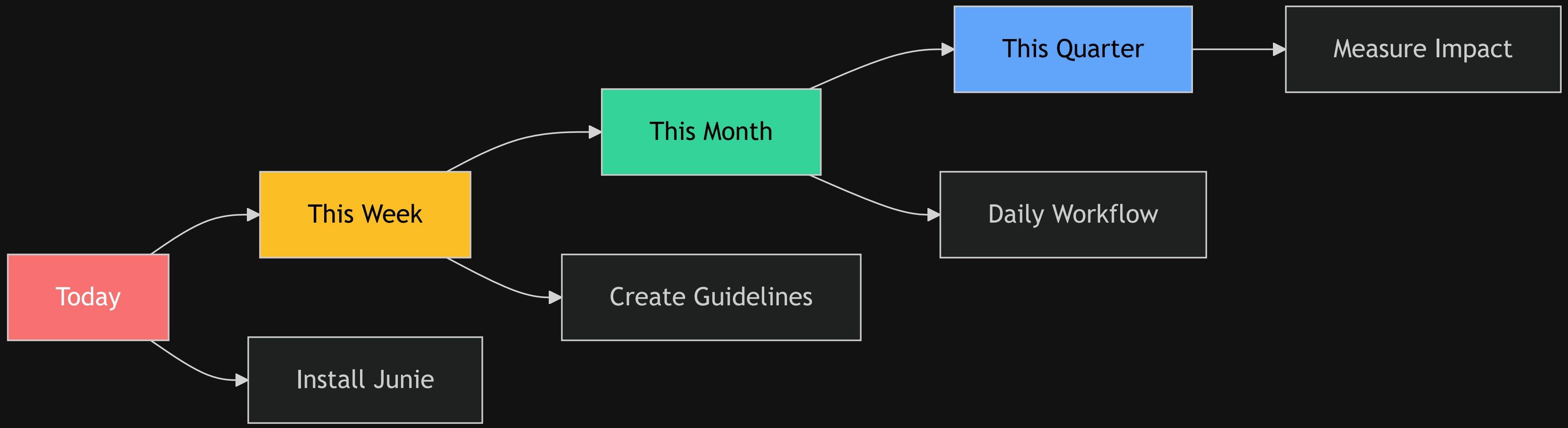
✅ **Guidelines drive consistency** - Encode your team's DNA

✅ **Build trust gradually** - Approvals → Allowlist → Brave

✅ **MCP extends capabilities** - Leverage external tools

✅ **Safety first** - Review diffs, maintain control

🎯 Your Next Steps



🚀 Start with Approvals Mode → Build Trust → Scale Up

💬 Discussion Time

Let's Explore Together:

🤔 **What would you trust Junie to do unattended?**

🛡️ **Where do you always want human review?**

📋 **How could guidelines help your team?**

🔧 **Which MCP tools fit your workflow?**

🚀 **How will AI agents change development?**

📚 Resources Hub

Documentation

- 📖 Getting Started
- 📝 Guidelines Guide
- 🔧 MCP Settings

Tutorials

- 💡 IntelliJ Playbook
- 🐍 PyCharm Playbook

GitHub

- 🗂️ Guidelines Catalog
- 📚 Context7 MCP
- 🧪 This repository

Community

- 💬 JetBrains AI Slack
- 🏷️ Stack Overflow: `jetbrains-junie`
- 📧 Support team

🙏 Thank You!

AI amplifies good practices

💡 Remember:
Invest in your guidelines today,
reap consistency forever!

Questions? Let's explore together! 🚀

About Ken Kousen

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AI agents aren't here to replace developers—
they're here to make us **better** developers.

🎁 Bonus: Quick Reference

Keyboard Shortcuts

- `Ctrl+Alt+J` - Open Junie
- `Ctrl+Enter` - Execute
- `Esc` - Cancel operation
- `Ctrl+Z` - Undo changes

Common Commands

"Analyze this file"

"Add tests with 90% coverage"

"Refactor to best practices"

"Fix all type errors"

MCP Tools

```
{
  "context7": "Library docs",
  "playwright": "E2E tests",
  "search": "Web search",
  "custom": "Your tools"
}
```

🎯 Print this slide for your desk!